Safety Data Sheet

According to Model Work Health and Safety Regulations 2011, Model Code of Practice for the Labelling of Workplace Hazardous Chemicals (2011), Guidance on the Classification of Hazardous Chemicals under the WHS Regulations (2012) (GHS 2009)

Triethanolamine 99%

Version 1.3 Release Date: 11.10.2022

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name	:	Triethanolamine 99%
Synonyms	:	Tri(2-Hydroxyethyl)Amine, TEA, TEA 99
CAS-No.	:	102-71-6

Recommended use of the chemical and restrictions on use

Recommended use	:	Use in personal care and construction industry
Restrictions on use	:	No restriction of use

Manufacturer or supplier's details

Headquarters

Company	:	PETRONAS Chemicals Group Berhad
Address	:	Tower 1, PETRONAS Twin Towers, Kuala Lumpur City Centre, 50088 Kuala Lumpur Malaysia

Plant Site

Company	:	PETRONAS Chemicals Derivatives Sdn Bhd
Address	:	Administration Complex, Kerteh Industrial Area, KM 106 Jalan Kuala Terengganu - Kuantan, 24300 Kerteh, Kemaman, Terengganu, Malaysia
Emergency telephone number	:	24-Hour Emergency Contact: +61 2 9037 2994

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Skin corrosion/irritation	:	Category 2
Serious eye damage/eye irritation	:	Category 2A
Specific target organ toxicity - single exposure	:	Category 3 (Respiratory system)

GHS label elements

Hazard pictograms :

Signal word	:	Warning
Hazard statements	:	H315 Causes skin irritation. H319 Causes serious eye irritation. H335 May cause respiratory irritation.
Precautionary statements	:	**Prevention:** P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray. P264 Wash skin thoroughly after handling. P271 Use only outdoors or in a well-ventilated area. P280 Wear protective gloves/ eye protection/ face protection.

Response:
P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P304 + P340 + P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P332 + P313 If skin irritation occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P362 Take off contaminated clothing and wash before reuse.
Storage:
P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

No information available.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture	:	Substance
Chemical nature	:	Ethanolamine
Substance name	:	Tri(2-Hydroxyethyl)Amine, TEA, TEA 99
CAS-No.	:	102-71-6

Components

Chemical name	CAS-No.	Concentration (%)
Triethanolamine	102-71-6	>= 99 -<= 100

SECTION 4. FIRST AID MEASURES

If inhaled	:	If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a medical doctor. Move to fresh air.
In case of skin contact	:	If on skin, rinse well with water. If on clothes, remove clothes. If skin irritation persists, call a medical doctor.
In case of eye contact	:	In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Continue rinsing eyes during transport to hospital. Remove contact lenses. Protect unharmed eye.

		Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Keep respiratory tract clear. Do NOT induce vomiting. If patient is fully conscious, give two glasses of water. Do NOT give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Take victim immediately to hospital. If symptoms persist, call a medical doctor. Clean mouth with water and drink afterwards plenty of water. Induce vomiting immediately and call a medical doctor.
General advice	:	Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
Most important symptoms and effects, both acute and delayed	:	No information available.
Notes to physician	:	If burn is present, treat as any thermal burn, after decontamination. No specific antidote. Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient.

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Water fog or fine spray. Dry chemical fire extinguishers. Carbon dioxide fire extinguishers. Foam. Alcohol resistant foams (ATC type) are preferred. General purpose synthetic foams (including AFFF) or protein foams may function, but will be less effective.
Unsuitable extinguishing media	:	High volume water jet Do not use direct water stream. May spread fire.
Specific hazards during firefighting	:	Do not allow run-off from fire fighting to enter drains or water courses.
Specific extinguishing methods	:	Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed off in accordance with local regulations.
Special protective equipment for firefighters	:	Wear positive-pressure self-contained breathing apparatus (SCBA) and protective fire fighting clothing (includes fire fighting helmet, coat, trousers, boots, and gloves). Avoid contact with this material during fire fighting operations. If contact is likely, change to full chemical resistant fire fighting clothing with self-contained breathing apparatus. If this is not available, wear full chemical resistant clothing with self-contained breathing apparatus and fight fire from a remote location. If protective equipment is not available or not used, fight fire from a protected location or safe distance. For protective equipment in post-fire or non-fire cleanup situations, refer to the relevant sections.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Environmental precautions	:	Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	:	Small spills: Absorb with materials such as: Non-combustible material. Sand. Clay. Vermiculite. Zorb-all®. Collect in suitable and properly labeled containers. Large spills: Contain spilled material if possible. Wipe up with absorbent material (e.g. cloth, fleece). Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Normal measures for preventive fire protection.
Advice on safe handling	:	Avoid formation of aerosol. Do not breathe vapours/dust. Provide sufficient air exchange and/or exhaust in work rooms. To avoid spills during handling keep bottle on a metal tray. Do not get in eyes, on skin, on clothing. Wash thoroughly after handling. Keep container closed. Use with adequate ventilation. Do not add nitrites or other nitrosating agents. Suspected cancer-causing nitrosamines could be formed. Spills of these organic materials on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion. Avoid contact with skin and eyes.
Hygiene measures	:	Wash hands before breaks and at the end of workday. When using do not eat or drink. When using do not smoke.
Conditions for safe storage	:	Keep container tightly closed in a dry and well-ventilated place. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards. Store in a dry place. Avoid freezing. Thaw and mix well before using. Store in the following material(s): Stainless steel. Do not store in: Galvanized steel. Copper. Copper alloys. Zinc.
Materials to avoid	:	No materials to be especially mentioned.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type	Control	Basis

		(Form of exposure)	parameters / Permissible concentration	
Triethanolamine	102-71-6	TWA	5 mg/m3	AU OEL
	Further information: Sensitiser			
		TWA	5 mg/m3	ACGIH

Personal protective equipment

Respiratory protection : Atmospheric levels should be maintained below the exposure guideline. For most conditions, no respiratory protection should be needed; however, if material is heated or sprayed, use an approved air-purifying respirator. Use a CE approved air-purifying respirator with cartridge/filter for: Organic vapor cartridge with a particulate pre-filter, type AP2.
In the case of vapour formation use a respirator with an approved filter.

Hand protection

Remarks : Use chemical resistant gloves classified under standard EN 374:
Protective gloves against chemicals and micro-organisms. Examples of preferred glove barrier materials include: Natural rubber ("latex"). Neoprene. Polyethylene. Ethyl vinyl alcohol laminate ("EVAL"). Polyvinyl alcohol ("PVA"). Polyvinyl chloride ("PVC" or "vinyl"). Examples of acceptable glove barrier materials include: Nitrile/butadiene rubber ("nitrile" or "NBR"). When prolonged or frequently repeated contact may occur, a glove with a protection class of 4 or higher (breakthrough time greater than 120 minutes according to EN 374) is recommended. When only brief contact is expected, a glove with a protection class of 1 or higher (breakthrough time greater than 10 minutes according to EN 374) is recommended.
The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash fountain should be located in immediate work area.
Use chemical goggles.
Chemical goggles should be consistent with EN 166 or equivalent.
Wear face-shield and protective suit for abnormal processing problems.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Selection of specific items such as face shield, gloves, boots, apron or full body-suit will depend on operation.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : Liquid
Colour : Colourless
Odour : Ammoniacal
Odour Threshold : No data available
pH : 8.0

Melting point/freezing point	:	> 20.0 °C
Boiling point/boiling range	:	335.4 °C
Flash point	:	185 °C
Evaporation rate	:	No data available
Upper explosion limit / Upper flammability limit	:	Not applicable
Lower explosion limit / Lower flammability limit	:	Not applicable
Vapour pressure	:	< 0.013 hPa
Relative vapour density	:	5.1(Air = 1.0)
Relative density	:	1.126 (15 °C)
Density	:	1.126 g/cm3 (15 °C) Method: ASTM D 4052
Solubility(ies)		
Water solubility	:	soluble
Partition coefficient: n-octanol/water	:	log Pow: -1.00
Auto-ignition temperature	:	324 °C
Viscosity		
Viscosity, dynamic	:	313.5 mPa.s (20 °C)Method: ASTM D 445
Viscosity, kinematic	:	182.4 mm2/s (40 °C)Method: ASTM D 445
Molecular weight	:	149.19 g/mol

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	Hazardous polymerisation does not occur.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	No dangerous reaction known under conditions of normal use.
Incompatible materials	:	Avoid contact with: Nitrites. Strong acids. Strong oxidizers. Product may potentially react with various halogenated organic solvents, resulting in temperature and/or pressure increases. Heating above 60°C in the presence of aluminum can result in corrosion and generation of flammable hydrogen gas. Avoid unintended contact with: Halogenated hydrocarbons. Corrosive when wet.
Hazardous decomposition products	:	Fumes, smoke, carbon monoxide Decomposition products depend upon temperature, air supply and the presence of other materials.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

<u>Components:</u>
Triethanolamine:

Acute oral toxicity	:	LD50 (Rat): 6,400 mg/kg
Acute inhalation toxicity	:	Remarks: No data available
Acute dermal toxicity	:	LD50 (Rabbit): > 2,000 mg/kg

Skin corrosion/irritation

Components:
Triethanolamine:

Species	:	Human skin
Result	:	Skin irritation

Serious eye damage/eye irritation

Components:
Triethanolamine:

Species	:	Rabbit
Result	:	Eye irritation

Respiratory or skin sensitisation

Components:
Triethanolamine:

Exposure routes	:	Inhalation
Remarks	:	No data available
Exposure routes	:	Skin contact
Result	:	Did not cause sensitisation on laboratory animals.

Chronic toxicity

Germ cell mutagenicity

Components:
Triethanolamine:

Germ cell mutagenicity - Assessment	:	In vitro tests did not show mutagenic effects

Carcinogenicity

Components:
Triethanolamine:

Carcinogenicity - Assessment	:	No data available

Reproductive toxicity

Components:
Triethanolamine:

Reproductive toxicity - Assessment	:	No data available

STOT - single exposure

Components:
Triethanolamine:

Target Organs	:	Respiratory tract irritation
Assessment	:	The substance or mixture is classified as specific target organ toxicant, single exposure, category 3 with respiratory tract

irritation.
The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Components:
Triethanolamine:
Remarks : No data available

Aspiration toxicity

Components:
Triethanolamine:
Statement on Aspiration Tox. : No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:
Triethanolamine:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 11,800 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Ceriodaphnia dubia (Water flea)): 609.88 mg/l Exposure time: 48 h
Toxicity to algae	:	ErC50 (Desmodesmus subspicatus (green algae)): 216 mg/l Exposure time: 72 h
Toxicity to fish (Chronic toxicity)	:	Remarks: No data available
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 16 mg/l Exposure time: 21 d
Toxicity to microorganisms	:	IC50 (activated sludge): > 1,000 mg/l Exposure time: 3 h

Persistence and degradability

Components:
Triethanolamine:
Biodegradability : Result: Readily biodegradable.

Bioaccumulative potential

Components:
Triethanolamine:
Bioaccumulation : Bioconcentration factor (BCF): < 3.9
Partition coefficient: n-octanol/water : Pow: 1.00

Mobility in soil

Components:
Triethanolamine:
Mobility : Medium: Soil
Remarks: No data available

Other adverse effects

Product:

Additional ecological information	:	No data available

Components:

Triethanolamine:

Additional ecological information	:	No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	Do not dispose of waste into sewer. Do not contaminate ponds, waterways or ditches with chemical or used container. For proper disposal of used material, an assessment must be completed to determine the proper and permissible waste management options permissible under applicable rules, regulations and/or laws governing your location. It may be feasible to flush a small spill of ethanolamines to a sanitary sewer, with large amounts of water. However, a large spill might be detrimental to aquatic life. If spilled material cannot be collected, it may be possible to neutralize with dilute hydrochloric acid and then, dispose of the resulting salt in accordance with national and local regulations. Can be landfilled after concentration, when in compliance with local regulations.
Contaminated packaging	:	Empty containers can only be disposed of when the remaining waste products adhering to the container walls have been removed. Hazard warning labels should be removed from the container walls. Incinerate in a furnace where permitted under national and local regulations. Dispose in accordance with all national and local environmental regulations. Empty containers should be recycled or disposed of through an approved waste management facility. Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Disposal methods identified are for the product as sold. For proper disposal of used materials, an assessment must be completed to determine the proper and permissible waste management options permissible under applicable rules regulations and/or laws governing your location.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG
Not regulated as a dangerous good

IATA-DGR
Not regulated as a dangerous good

IMDG-Code
Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollution category	:	Z
Ship type	:	3

National Regulations

ADG
Not regulated as a dangerous good

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Standard for the Uniform Scheduling of Medicines and Poisons	:	Schedule 5

The components of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory.
TSCA	:	All substances listed as active on the TSCA inventory
AICS	:	On the inventory, or in compliance with the inventory.
DSL	:	All components of this product are on the Canadian DSL
ENCS	:	On the inventory, or in compliance with the inventory.
ISHL	:	On the inventory, or in compliance with the inventory.
KECI	:	On the inventory, or in compliance with the inventory.
PICCS	:	On the inventory, or in compliance with the inventory.
IECSC	:	On the inventory, or in compliance with the inventory.
NZIoC	:	Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

Revision Date	:	11.10.2022
Sources of key data used to compile the Safety Data Sheet	:	ECHA - European Chemicals Agency
		JAPAN NITE

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists

AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
C	-	Ceiling
CCHC	-	Chemicals Classification and Hazard Communication
CEIL	-	Ceiling
ChV	-	Chronic Toxicity Value
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System of Classification and Labelling of Chemicals
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSPP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature

SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System
LLx	-	Loading rate associated with x% lethality effect

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